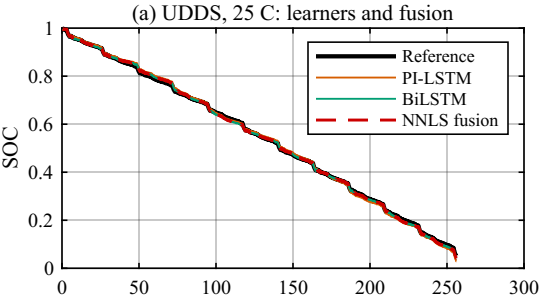
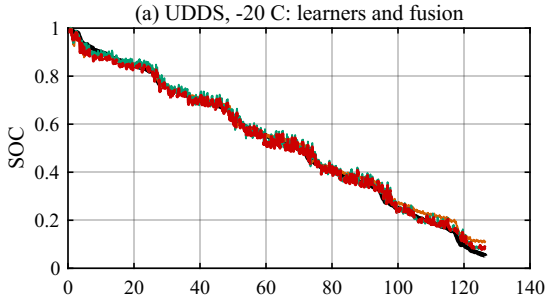
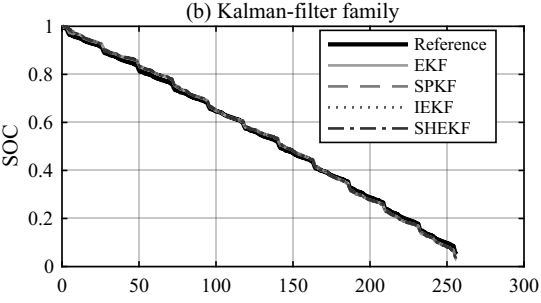
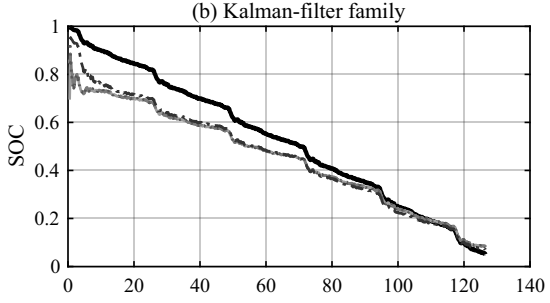
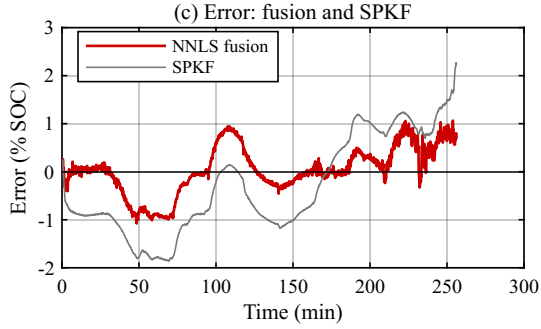
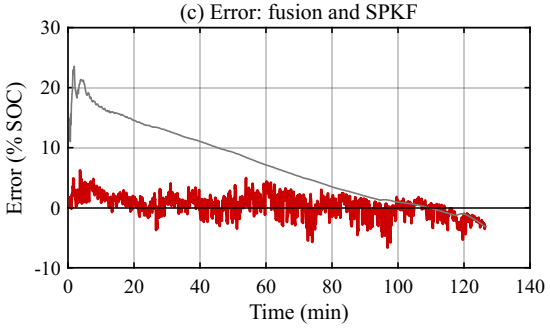

(a) UDDS, 25 C: learners and fusion
SOC
Reference
PI-LSTM
BiLSTM
NNLS fusion


(a) UDDS, -20 C: learners and fusion
SOC


(b) Kalman-filter family
SOC
Reference
EKF
SPKF
IEKF
SHEKF


(b) Kalman-filter family
SOC


(c) Error: fusion and SPKF
Error (% SOC)
NNLS fusion
SPKF
Time (min)


(c) Error: fusion and SPKF
Error (% SOC)
Time (min)